ECE 285 Homework 01

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1 Method

We implement a convolutional variational autoencoder (VAE) in PyTorch and train it on the Kaggle *Chest X-Ray Images (Pneumonia)* dataset to generate 64×64 chest radiographs. For $x \in [0,1]^{1\times64\times64}$ we use a diagonal Gaussian posterior $q_\phi(z \mid x) = \mathcal{N}(\mu_\phi(x), \operatorname{diag}\sigma^2_\phi(x))$, a factorised Bernoulli likelihood and a standard normal prior. The β-weighted negative ELBO per image, in nats, is

$$\mathcal{L} = \underbrace{\mathbb{E}_{q_\phi(z|x)}[-\log p_\theta(x \mid z)]}_{\text{BCE, summed over pixels}} + \beta \underbrace{D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, \mathcal{N}(0, I))}_{\text{closed form}}, \qquad z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon, \quad (1)$$

where $D_{\mathrm{KL}} = -\frac{1}{2}\sum_j (1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2)$ and the BCE is summed over pixels then divided by the batch size. Encoder and decoder are mirror-image convolutional stacks of $\log_2 64 - 2 = 4$ stages (4×4 kernels, stride 2, widths $32 \to 64 \to 128 \to 256$); two linear heads give $(\mu, \log\sigma^2) \in \mathbb{R}^{128}$, and the decoder starts from $\mathrm{FC}(128 \to 4096)$ and ends in a sigmoid. Batch-norm is omitted on the last encoder and first decoder block. Training uses Adam with 10^{-3} cosine-decayed to 10^{-5} and gradient-norm clipping at 5.0; β ramps from 0 to 1 over 10 epochs, the standard remedy for early posterior collapse. Table 1 lists the configuration.

Table 1: Hyperparameter settings.

image size	64×64, 1 ch.	batch size	64
latent dim. d	128	parameters	2,955,905
train / val / test	4,694 / 522 / 624	real reference	512 images
optimiser	Adam (0.9, 0.999)	learning rate	$10^{-3} \to 10^{-5}$
epochs	50 (73 iters)	KL weight β	$0 \to 1$ over 10
grad. clipping	norm 5.0	evaluation	every 5 epochs

2 Experimental setup

The dataset holds 5,863 AP/PA radiographs, split officially into 5,216 / 16 / 624. The official validation split has only 16 images, so we discard it and take a class-stratified 90/10 split of the official training set (4,694 / 522, preserving the 74.3 % pneumonia prevalence). The 624-image test set is never optimised on; a fixed 512-image subset is the *real* reference. Images are single-channel, resized to 64×64 with anti-aliased bilinear interpolation and scaled to $[0, 1]$ since the decoder predicts a Bernoulli mean; we apply *no* horizontal flips because chest radiographs are left/right asymmetric. For the Inception metrics a second pipeline produces $3 \times 299 \times 299$ tensors from the *original* JPEGs, giving a sharper reference than upsampled crops.

3 Results

Training dynamics. The loss (Fig. 1) falls from 2728.97 to 2388.75 nats per image — 0.583 nats/pixel — and still decays marginally at the end. Validation tracks training closely (best 2362.41 at epoch 49), i.e. no overfitting, as expected for a 128-d bottleneck on 4,694 examples. The KL term *contracts* from 112.38 to 24.08 nats while β rises from 0.1 to 1.0, the intended β-VAE trade-off: reconstruction improves fast enough that the encoder can approach the prior. The KL does not

collapse, but 24.08 nats $\approx$ 34.7 bits over 128 dimensions means few units are active — which explains the blurriness, since the code carries $\sim$35 bits about a 4,096-pixel image whose reconstruction term costs 3,412 bits.

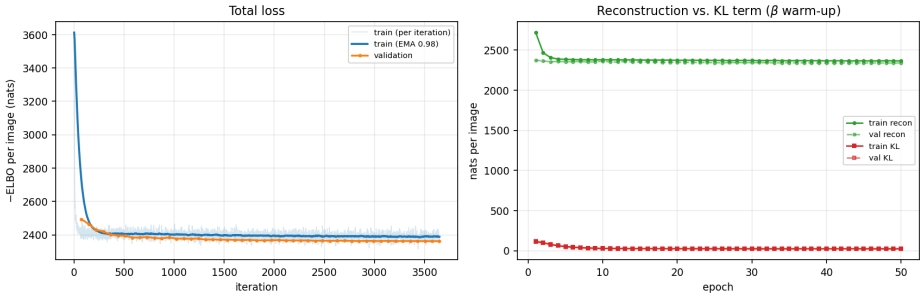


Figure 1: Training loss. **Left:** per-iteration loss (light) with its EMA (dark) and per-epoch validation loss. **Right:** reconstruction and KL terms during the β warm-up; KL contracts from 112.4 to 24.1 nats while β rises to 1.

Inception Score and FID. Both metrics use an ImageNet-pretrained InceptionV3, following the `pytorch-ignite` reference implementation: IS on its 1000-d softmax output and FID on its 2048-d pooling features, with inputs replicated to three channels and resized to 299×299 by PIL bilinear interpolation. At the best checkpoint (epoch 45) the model reaches **IS 1.618** and **FID 343.51**, against **2.289** and **66.30** for real images, i.e. 71 % of the real IS (Table 2). Over epochs 5–50, IS climbs from 1.283 to a plateau near 1.65 (peak 1.673 at epoch 45) while FID falls from 348.91 to 341.49 then rises to 344.39 at epoch 50. The two decouple late in training: IS keeps improving as samples become smoother and more prototypical, whereas FID compares full feature covariances and saturates earlier. After epoch 30 FID has mean 343.3 and standard deviation 1.7, so the 0.35 margin of epoch 45 over epoch 30 is not meaningful.

Table 2: FID and Inception Score on 512 held-out test images.

	Inception Score ↑	FID ↓
Real test images (reference)	2.289	66.30
VAE, this work (epoch 45)	1.618	343.51

Generated images. Prior samples (Fig. 2, left) are unmistakably chest radiographs: rib cage, lung fields, mediastinal column and diaphragm are present, with visible variation in lung width, heart shadow and exposure. Missing is high-frequency detail — ribs appear as soft bands and no vascular markings survive. This is a property of the objective rather than of the optimisation: under a Bernoulli likelihood the optimal prediction is the conditional mean of each pixel given z, so averaging over residual uncertainty necessarily yields a smooth image. Reconstructions recover global geometry at MSE 0.0034 while losing the same fine structure, consistent with the 24-nat bottleneck.

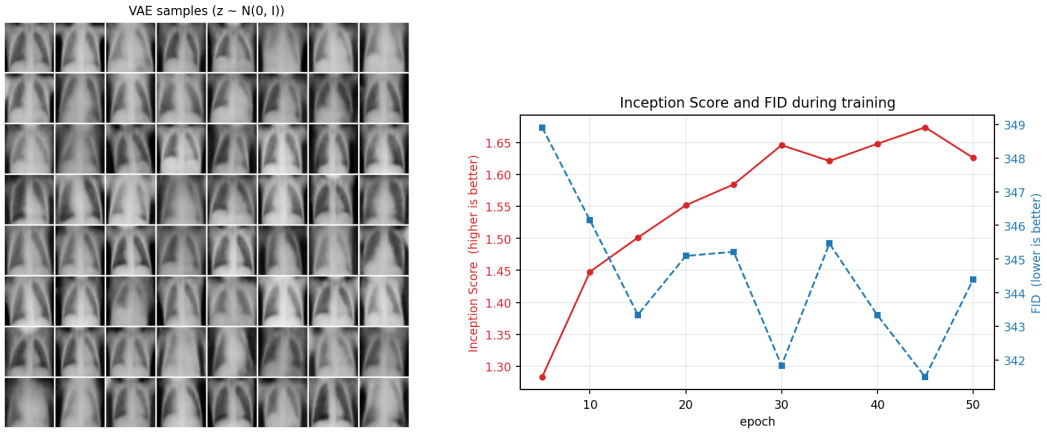


Figure 2: Left: 64 samples from the prior $z \sim \mathcal{N}(0, I)$ at epoch 45. **Right:** IS (higher better) and FID (lower better) every 5 epochs on the fixed 512-image reference set.